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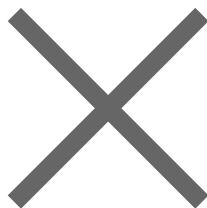
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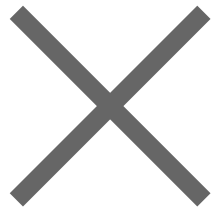
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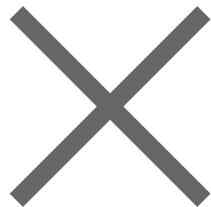
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Co-packaged optics-based networking switches with unmatched power efficiency and resiliency.

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Introduction

Revolutionizing Networking for the Era of Agentic AI

NVIDIA's co-packaged optics (CPO) switches with integrated silicon photonics are the world's most advanced networking solution for the era of agentic AI. Replacing pluggable transceivers with silicon photonics on the same package as the ASIC, NVIDIA CPO innovations provide 5x better power efficiency and higher network resiliency resulting in 5x sustained AI application runtime.

NVIDIA CPO-based networks simplify manageability and design and enable more efficient-compute infrastructure. These benefits are critical to delivering the scale needed to enter the future of million-GPU [AI factories](#).

Spectrum-X Photonics for Vera Rubin Enable Million-GPU AI Factories

Design simplicity and power efficiency drives CoreWeave, Lambda, Meta, Microsoft, and Oracle Cloud Infrastructure as first adopters.

[Read Press Release](#)

Spectrum-X Ethernet Photonics Ramps to Full Production

With NVIDIA full-stack codesign, Spectrum-X™ Ethernet Photonics reaches production through deep co-engineering with the semiconductor and systems ecosystem.

[Read Blog](#)

NVIDIA Co-Packaged Optics Explained

Take a look inside the new NVIDIA silicon photonics-based networking switches.

[Watch Video](#)

Offerings

NVIDIA CPO Architectures

NVIDIA Quantum-X InfiniBand Photonics

Co-packaged optics built on 200G SerDes technology for the NVIDIA Quantum-X800 InfiniBand platform.

NVIDIA Spectrum-X Ethernet Photonics

Co-packaged optics built on 200G SerDes technology for the NVIDIA Spectrum-X™ Ethernet platform.

Benefits

Networking for Massive-Scale Infrastructure

NVIDIA CPO introduces many benefits that are necessary to scale to millions of GPU AI factories.

Reduced Power Consumption

NVIDIA CPO technology significantly reduces network power, enabling **5x** better power efficiency over pluggable transceivers.

Higher Resiliency

NVIDIA CPO delivers higher resiliency over pluggable transceivers resulting in **5x** sustained AI application runtime.

Faster Deployment

NVIDIA CPO simplifies data center networking installation and serviceability, leading to **1.3x** faster time to insight.

Lower Latency

NVIDIA CPO doesn't require digital signal processing (DSP) retimers, reducing network latency.

Lower TCO

NVIDIA CPO eliminates the need for pluggable transceivers, simplifying the bill of materials and reducing cost.

Easier Serviceability

NVIDIA CPO requires fewer components and is much easier to install and replace than pluggable transceivers.

NVIDIA Silicon Photonics Switches

NVIDIA Quantum-X InfiniBand Photonics Switch

The NVIDIA Quantum-X800 InfiniBand platform includes CPO-based switches, including the Q3450-LD with 144 ports of 800 gigabits-per-second (Gb/s) InfiniBand. This groundbreaking switch leverages a liquid-cooled design to efficiently cool the onboard silicon photonics. The NVIDIA Quantum-X InfiniBand Photonics switch supports network innovations that deliver unprecedented scale, connecting over 10,000 GPUs in a non-blocking, two-level fat-tree topology.

[Learn More About the Quantum-X InfiniBand Photonics Switch](#)

NVIDIA Spectrum-X Ethernet Photonics Switches

NVIDIA's Spectrum-X Ethernet Photonics switches usher in the next era of AI infrastructure by integrating co-packaged optics (CPO) directly onto the ASIC, overcoming the limits of electrical signaling in large-scale AI factories. With up to 409.6 terabits-per-second (Tb/s) bandwidth, Spectrum-X Ethernet Photonics enables massive generative AI workloads. Available in the second half of 2026, it sets a new standard for scalable, energy-efficient Ethernet networks that power million-GPU AI clusters.

[Learn More About the Spectrum-X Ethernet Photonics Switch](#)

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1. Blogs
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A New Era in Data Center Networking With NVIDIA Silicon Photonics-Based Network Switching

Explore why co-packaged silicon photonics can accelerate large-scale AI model development and inference with benefits like lower power consumption, reduced latency, and network resiliency over traditional pluggable optical transceivers.

[Read Blog](#)

Scaling AI Factories With Co-Packaged Optics for Better Power Efficiency

NVIDIA Photonics is redefining data center networking with co-packaged optics that eliminate pluggable transceivers, boosting resiliency and power efficiency.

[Read Blog](#)

How Industry Collaboration Fosters NVIDIA Co-Packaged Optics

NVIDIA co-packaged optics solutions, built in collaboration with partners like TSMC, set the foundation for scalable, energy-efficient AI infrastructures of the future.

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Dive Into NVIDIA Co-Packaged Optics-Based Switches

Take a look inside NVIDIA silicon photonics-based networking switches that simplify manageability and design, enabling more power for compute infrastructure and delivering the scale needed to enter the future of million-GPU AI factories.

[Watch Video \(01:32\)](#)

Powering on NVIDIA Quantum-X InfiniBand Photonics Switch

Watch NVIDIA's Quantum-X InfiniBand Photonics switch come to life in an AI factory connecting NVIDIA's GB300 racks with OSFP pluggable optical modules demonstrating scale-out topology using NVIDIA Silicon Photonics, the world's most advanced networking solution for the era of agentic AI.

[Watch Video \(01:26\)](#)

Next Steps

Ready to Get Started?

Configure Your Cluster

This online tool can help you configure clusters based on fat tree with two levels of switch systems and Dragonfly+ topologies.

[Get Started](#)

Take Networking Courses

Explore deep technical training topics on NVIDIA Quantum InfiniBand networking through the NVIDIA Academy.

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Ready to Purchase?

Visit the NVIDIA marketplace to discover more information on how to purchase NVIDIA networking solutions.

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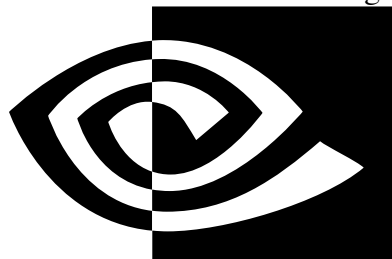
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